

TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Long-range terahertz vision sensor

US-based startup Teradar introduced its Teradar Summit, claiming it as the first long-range terahertz (THz) vision



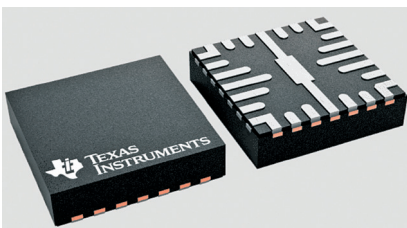
sensor that works in any weather. It is built to support driver assistance systems (ADAS) and L3-L5 autonomy. The Summit features a solid-state digital phased-array, offering a 300-metre range with 0.13° resolution. It produces 3D point clouds with Doppler data, capturing range, direction, elevation, and relative speed, supporting autonomous systems from L2 to L5. The technology fills gaps left by radar and lidar, enabling vehicles to perceive more accurately. Its chip offers up to 20x the resolution of automotive radar, supporting detailed all-weather perception.

Teradar

<https://teradar.com>

Single-chip 4D radar

Texas Instruments (TI) has launched the AWR2188 4D imaging radar transceiver, which lets automakers build radar systems with less hardware and



lower complexity by integrating eight transmitters and eight receivers on a single chip. This single-chip design supports both satellite and edge architectures, making it easier to add ADAS features across entry-level to premium vehicles. ADC data processing and a radar chirp signal slope engine make detection 30% faster than current solutions, allowing tracking of objects over 350 metres, even in crowded, high-dynamic, or cargo-obscured environments. By enabling higher channel counts with fewer devices, the AWR2188 helps engineers deploy radar capabilities efficiently and supports safer, more autonomous driving.

Texas Instruments (TI)

<https://www.ti.com>

Encoder MCU

General purpose MCUs often fall short in meeting the demands of next-generation encoders, so Geehy has introduced its first dedicated encoder



MCU, the G32R430, built for magnetic, optical, and inductive encoder applications. The device integrates analogue, digital, and power management functions on a single chip, simplifying system design and reducing PCB area and BOM costs. It includes a 12-bit ADC with up to 14 external and two internal channels, two 10-bit DACs that can be configured as comparator inputs, four programmable analogue comparators, a temperature sensor,

and main supply voltage sensing, removing the need for many external components. The G32R430 can reduce power consumption by about 50% compared to traditional solutions while improving real-time processing and accuracy.

Geehy Semiconductor

<https://global.geehy.com>

FirePower laser firing system ICs

Silanna Semiconductor announced that its FirePower laser firing system ICs are now available. The SL2001 and SL2002 reduce power use, system size,



bill of materials, and design steps compared with other laser driver solutions. These ICs combine resonant capacitor charging and high-current laser diode firing into a single device, eliminating the need for multiple external components and additional conversion stages. For example, using the SL2001 with a 400W quad-edge-emitting laser module can reduce a laser system's footprint by 80%. Both devices support resonant-capacitor charging efficiency up to 85% and work with edge-emitting lasers (EELs) and vertical-cavity surface-emitting lasers (VCSELs).

Silanna Semiconductor

<https://silannasemi.com>

4D lidar sensor

Aeva has introduced Aeva Omni, a 4D lidar sensor for physical AI applications. Omni works in autonomous